Magnetism of Dikes of the Frontenac Axis[1]

J. K. PARK AND E. IRVING
Earth Physics Branch, Department Energy Mines and Resources, Ottawa

Manuscript received March 17, 1972
Accepted for publication March 24, 1972

Paleomagnetic results from 11 dikes of three different directional trends (north, northwest, and northeast) from the Frontenac axis are reported. Six northwest trending dikes (minimum age 800 m.y.) give well-grouped stable magnetization directions with a corresponding pole at 12 °S, 162 °E ($N = 6$, $K = 85$, $A_{95} = 7°$).

On fait rapport ici d'études de paléomagnétisme, effectuées dans l'Axe de Frontenac, sur 11 dikes de trois orientations différentes (Nord, Nord-Ouest, et Nord-Est). Les six dikes d'orientation Nord-Ouest (âge minimum 800 m.a.) livrent des directions de magnétisation stables et bien groupées, avec un pôle correspondant à 12 °S, 162 °E ($N = 6$, $K = 85$, $A_{95} = 7°$).

The dikes of the Frontenac axis are not numerous and are often poorly exposed; the paleomagnetic results from them are not of especial interest until they can be integrated with observations from elsewhere in the Canadian Shield. For this reason the data are briefly described herein without interpretation. Sampling of these dikes was initiated by W. F. Fahrig of the Geological Survey of Canada and we are grateful to him for making his material available to us. We have extended the sampling and carried out the magnetic measurements.

The collecting sites are marked in Fig. 1. All dikes shown on the available geological maps were visited, but only 14 exposures were suitable for sampling. The other localities were either covered by vegetation or deeply weathered. No studies of contacts were made because of the generally poor exposure. Three dikes (7, 11, and 13) gave scattered directions even after demagnetization, and no useful information was retrieved from them. The results from the other localities are summarized in Table 1. The dikes may be grouped according to their directional trend.

Northwest-Trending Dikes

The 6 dikes sampled have moderate coercivities. The initial directions are widely scattered. During alternating field demagnetization some

TABLE 1. Summary of results

lat(N), long(W)	a.f. (Oe)	*D*,*I*	*N*	*k*	α_{95}	pole	m.d.f. of i.r.m.
NORTHWEST TREND							
1 44°16.7′, 76°17.8′	400	100, + 43	5	59	10	11S, 169E	170
2 44°16.0′, 76°27.2′	300	101, + 46	6	96	07	11S, 166E	162
3 44°16.7′, 76°17.1′	300	104, + 49	6	485	03	11S, 162E	143
5 44°17.2′, 76°12.7′	500	120, + 59	6	59	09	10S, 147E	157
6 44°17.7′, 76°13.1′	300	106, + 49	5	129	07	10S, 162E	162
10 44°16.6′, 75°13.8′	300	093, + 51	9	83	06	20S, 168E	298†
Average of directions	—	103, + 50	6	112	06	12S, 163E	—
Average of poles	—	—	6	85	07	12S, 162E	—
NORTHEAST TREND							
8 44°17.3′, 75°52.6′	200	111, + 70	8	9	20	24S, 141E	174
9 44°19.1′, 76°41.4′	300	139, + 45	6	7	28	10N, 140E	352†
12 44°19.1′, 75°52.6′	300	283, + 30	7	30	11	20N, 163W	152
15 44°17.3′, 76°22.3′	300	131, + 60	8	28	11	06S, 139E	233
NORTH TREND							
4* 44°16.7′, 76°26.3′	500	315, − 15	9	134	05	24N, 154E	156

The cleaning field in oersteds is given in the second column. *D*, *I* are the declination and inclination of the mean. *N* is the number of independent cores (2 specimens measured from each). *k* and α_{95} are Fisher's estimate of precision and circle of confidence (Fisher 1953). The coercivity of maximum isothermal remanent magnetization (i.r.m. acquired in 12 000 Oe) are given in the last two columns; there are the demagnetizing fields required to demagnetize one half of the initial remanence the procedures being explained in Park and Irving (1970). Site 4* is the average of results from sites 4 and 14 (Fig. 1) giving unit weight to cores ($N = 9$). Sites given unit weight ($N = 6$) for the "northwest average". †Average of two specimens.

[1]Earth Physics Branch Contribution no. 390.

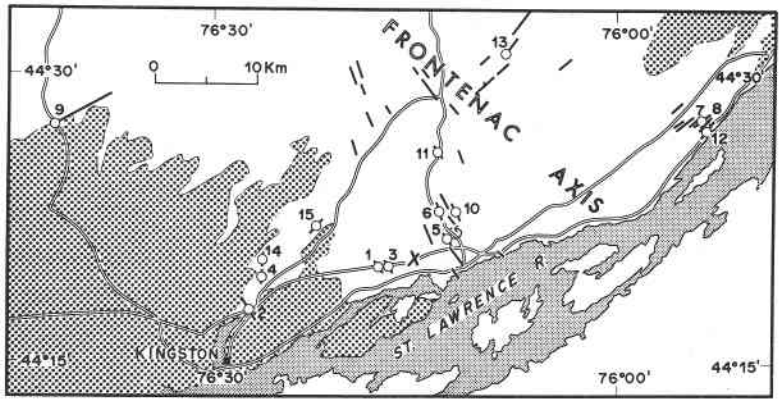


FIG. 1. Locality map. The blank area is Precambrian (Grenville Province) and the stippled area subhorizontal Lower Paleozoic sediments. The dikes are shown diagrammatically, their widths being grossly exaggerated. The site numbers are catalogue numbers and have no geological significance.

specimens show remarkable directional variations before reaching stable end-points; for example one specimen from site 1 shows a 180° reversal. After magnetic cleaning the agreement of direction, both within and between-sites, is excellent (Fig. 2 site 3, Table 1); the mean standard deviation within-sites being 8° and the between-site standard deviation 8°. Whole-rock and biotite K–Ar ages of 751 ± 75 and 817 ± 70 m.y. respectively have been reported from site 5 (Wanless *et al.* 1966, 1967). These are not significantly different and indicate a minimum age of about 800 m.y.

North-Trending Dike

This dike was sampled at two sites (4, 14). The coercivity (that is the median point of the coercive force spectrum) is in the range normally expected for diabase, but very high coercive force components are present and the direction in some specimens was little changed after demagnetization in a.f. of 1000 Oe. The within-site grouping after cleaning is excellent. No age dates are available, but the pole agrees with those for Franklin diabase dated at 675 m.y. (Fahrig *et al.* 1971).

Northeast-Trending Dikes

These dikes give much less satisfactory data. They have variable, sometimes high coercivity, but the within-site dispersion is high even after magnetic cleaning. Two directions occur. Dike 12, with a westerly direction, gives the best data (Fig. 2). Two K–Ar whole-rock ages of 407 ± 23 and 411 ± 23 m.y. have been obtained from this body (data of Shafiqullah (1972) quoted with permission). Dikes 8, 9,

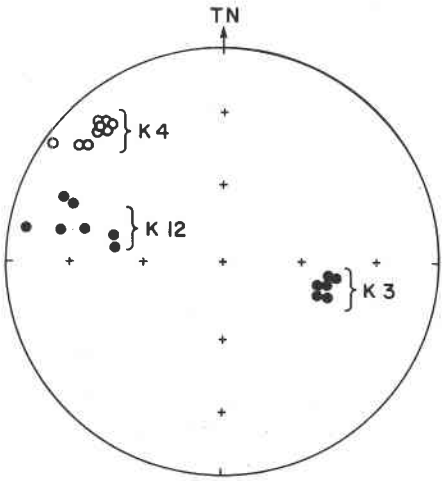


FIG. 2. Directions observed in cores (average of 2 specimens) from 3 sites after magnetic cleaning. Sites 3 (NW dikes) and 4 (N dike) show good grouping with very stable magnetization. Site 12 (NE dike) shows the greater scatter, characteristic of dikes of this trend.

and 15 give poor data with a SE direction not greatly different from that in the NW-trending dikes. A whole-rock K–Ar age of 434 ± 75 m.y. has been obtained from dike 15 (Wanless *et al.* 1966). The K–Ar dates suggest that these dikes are early Palaeozoic, but they are best regarded as minimum ages, especially since all are whole-rock determinations.

FAHRIG, W. F., IRVING, E., and JACKSON, G. D. 1971. Paleomagnetism of the Franklin diabases. Can. J. Earth Sci., **8**, pp. 455–467.

FISHER, R. A. 1953. Dispersion on a sphere. Proc. Roy. Soc. London, **217**, pp. 295–305.

PARK, J. K. and IRVING, E. 1970. The Mid-Atlantic Ridge near 45°N. XII. Coercivity, secondary magnetization, polarity and thermal stability of dredge samples. Can. J. Earth Sci., **7**, pp. 1499–1514.

SHAFIQULLAH, M. 1972. K–Ar ages on alkaline intrusions and dikes in Eastern Canada. In press.

WANLESS, R. K., STEVENS, R. D., LACHANCE, G. R., and EDMONDS, C. M. 1967. Age determinations and geological studies: K–Ar Isotopic ages, report 7. Geol. Surv. Can., Pap. 66-17.

WANLESS, R. K., STEVENS, R. D., LACHANCE, G. R., and RIMSAITE, J. Y. H. 1966. Age determinations and geological studies: K–Ar Isotopic ages, report 6. Geol. Surv. Can., Pap. 65-17.